HW#3

Homework Report

Course	ECG 762
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Problem 01 3.15

Independent bivariate Gaussian samples $\{\mathbf{x}[0], \mathbf{x}[1], \ldots, \mathbf{x}[N-1]\}$ are observed. Each observation is a 2×1 vector which is distributed as $\mathbf{x}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$ and $\mathbf{C} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$.

Find the CRLB for the correlation coefficient ρ. Hint: Use (3.32).

(3.32):

$$I(\theta) = \left[\frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta}\right]^T \mathbf{C}^{-1}(\theta) \left[\frac{\partial \boldsymbol{\mu}(\theta)}{\partial \theta}\right] + \frac{1}{2}\mathrm{tr}\left[\left(\mathbf{C}^{-1}(\theta)\frac{\partial \mathbf{C}(\theta)}{\partial \theta}\right)^2\right]$$

Solution:

Each observation is distributed as

$$x[n] \sim \mathcal{N}(0, C),\ C = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},\ |\rho| < 1.$$

Since the mean vector is independent of ρ,

$$\frac{\partial \mu}{\partial \rho} = 0.$$

Therefore, by (3.32), the Fisher information from a single observation is

$$I_1(\rho) = \frac{1}{2}\mathrm{tr}\left[\left(C^{-1}\frac{\partial C}{\partial \rho}\right)^2\right].$$

The inverse covariance matrix and its derivative are

$$C^{-1} = \frac{1}{1-\rho^2}\begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix},\ \frac{\partial C}{\partial \rho} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Thus,

$$C^{-1}\frac{\partial C}{\partial \rho} = \frac{1}{1-\rho^2}\begin{bmatrix} -\rho & 1 \\ 1 & -\rho \end{bmatrix},$$

and

$$\left(C^{-1}\frac{\partial C}{\partial \rho}\right)^2 = \frac{1}{(1-\rho^2)^2}\begin{bmatrix} 1+\rho^2 & -2\rho \\ -2\rho & 1+\rho^2 \end{bmatrix}.$$

Taking the trace gives

$$I_1(\rho) = \frac{1}{2}\frac{2(1+\rho^2)}{(1-\rho^2)^2} = \frac{1+\rho^2}{(1-\rho^2)^2}.$$

Because the Nobservations are independent, their Fisher information adds:

$$I_N(\rho) = NI_1(\rho) = \frac{N(1+\rho^2)}{(1-\rho^2)^2}.$$

Hence, the CRLB for any unbiased estimator $\hat{\rho}$is

$$\text{var}(\hat{\rho}) \geq \frac{1}{I_N(\rho)} = \frac{(1-\rho^2)^2}{N(1+\rho^2)}, |\rho| < 1.$$

Problem 02 3.19

A line array of antennas with $d = \lambda/2$ spacing for a 1 MHz electromagnetic signal is to be designed to yield a 5° standard deviation at $\beta = 90°$. If the SNR η is 0 dB, comment on the feasibility of the requirement. Assume that the line array is to be mounted on the wing of an aircraft. Use $c = 3 \times 10^8$ m/s.

Solution:

Let Mdenote the number of antenna elements. From the bearing estimation result in Example 3.15, equation (3.43), the CRLB is

$$\text{var}(\hat{\beta}) \geq \frac{12}{\eta M(M^2-1)\left(\frac{2\pi d}{\lambda}\right)^2 \sin^2\beta},$$

where $\beta$is measured in radians and $\eta$is the linear SNR.
For a 1 MHzelectromagnetic signal, the wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m}.$$

Therefore, the required element spacing is

$$d = \frac{\lambda}{2} = 150 \text{ m}.$$

Since

$$\eta = 10^{0/10} = 1,\ \sin^2 90° = 1,\ \frac{2\pi d}{\lambda} = \pi,$$

the bound reduces to

$$\text{var}(\hat{\beta}) \geq \frac{12}{\pi^2 M(M^2-1)}.$$

The desired standard deviation, expressed in radians, is

$$\sigma_{\text{target}} = 5\frac{\pi}{180} = \frac{\pi}{36} \approx 0.08727 \text{ rad}.$$

A necessary condition for meeting this requirement is that the CRLB does not exceed the target variance:

$$\frac{12}{\pi^2 M(M^2-1)} \leq \left(\frac{\pi}{36}\right)^2.$$

Rearranging gives

$$M(M^2-1) \geq \frac{15552}{\pi^4} \approx 159.66.$$

For $M = 5$,

$$M(M^2-1) = 5(25-1) = 120,$$

which does not satisfy the condition. For $M = 6$,

$$M(M^2-1) = 6(36-1) = 210,$$

which satisfies it.
Thus, the CRLB calculation requires at least six antenna elements. The corresponding array length is

$$L = (M-1)d = (6-1)(150) = 750 \text{ m}.$$

The requirement is therefore impractical for an array mounted on an aircraft wing: even the idealized CRLB calculation requires a 750 marray. Moreover, satisfying the CRLB condition is necessary but does not guarantee that a practical estimator will achieve the desired standard deviation.